

AUTHBC: Feasibility Boundaries for Authenticated UAV Telemetry — An Exclusion Bound and a Capacity Envelope

Mohamed A. Farouk

[AFFILIATION – TO BE COMPLETED]

[Department, Institution, City, Country]

mohamed.besco1998@gmail.com

Abstract—A UAV that broadcasts signed, hash-chained telemetry spends more bytes on authentication than on data: a 64 B signature and a 32 B chain link accompany a 24.0 B record. AUTHBC makes the frame the unit of design: a format, a sender and a receiver in which one signature and one chain link cover b records and every frame decodes and verifies alone. Batch size, encoding and signature scheme are chosen together under freshness, verifiability, channel capacity and receiver CPU; batching gains nothing unless record rate times deadline exceeds two. Where the frame does not fit: when every frame must verify alone, eight of twelve EU863-870 LoRaWAN data rates cannot carry one (five cannot hold the signature in any format, three our frame). Where it does: at 50 records/s and a 100 ms deadline the design sends 43.25 B per record against 145.77 B for per-record signing, and in ns-3 one 802.11a collision domain delivers 95% of frames among 124 nodes instead of 35. Sender energy is metered. Capacities are simulated; contention was measured on two radios. Record sizes are synthetic, checked against twelve flight logs and a simulated autopilot.

Index Terms—UAV telemetry, broadcast authentication, digital signatures, hash chain, batching, IEEE 802.11 DCF, LoRaWAN, ns-3.

I. INTRODUCTION

Unmanned aircraft in a swarm tell each other where they are many times a second. If those reports are to be evidence afterwards — of who was where, and of what each vehicle claimed — each one must be attributable to its sender and the sequence must be tamper-evident. The textbook construction signs every record and links it to its predecessor by a hash. The textbook cost is that the protection is larger than the thing protected: an Ed25519 or ECDSA signature is 64 B and a SHA-256 link is 32 B, while a compactly encoded position, velocity and status record is about 24.0 B. On a shared radio channel those bytes are airtime, and airtime decides how many neighbours can be heard.

Practice today signs per message. MAVLink 2 appends a 13 B symmetric tag to each message [1]; vehicular standards attach an ECDSA signature, and periodically a certificate, to each one [2]. The obvious alternative is to let several records share one frame, one signature and one chain link. Batching is not new, and its price is known in kind: records wait for the frame to fill, and a lost frame loses all of them. What we did not find is the accounting. What must a frame carry so that a receiver can verify it without any other frame? On which links

does such a frame not fit, whatever the batch? Where it fits, which constraint caps the batch, and what do the saved bytes buy on a contended channel, in nodes and not in percent? And can that be known without simulating every configuration?

We answer these for one concrete system. AUTHBC — authenticated telemetry for a blockchain-style, hash-chained ledger — is a frame format, a sender and a receiver. We built it, measured it, and searched its parameters. Our contributions are:

- **A capacity rule with its mechanism.** Capacity is found by simulating each configuration at its own frame size and rate, with 30 seeds and a bootstrap interval, not by applying one load ceiling to all. A rule with one fitted constant, the capacity envelope of the title, reproduces the simulated capacities. The access rule accounts for the constant and, with one loss mechanism that earlier analyses of broadcast do not contain, gives the capacities with nothing fitted. Every prediction made from either was written down before its runs, other batch sizes and unequal received power among them (Sec. VIII-D).
- **A ladder that separates causes.** Sharing a frame, sharing a signature and delta coding are separate steps, and we report bytes, capacity and receiver CPU for each (Sec. VIII), and sender energy with and without batching. Most of the capacity is gained by sending fewer frames, the effect frame aggregation exists for.
- **An exclusion bound with its scope.** On LoRaWAN, eight of twelve EU863-870 data rates cannot carry one signed, chained frame: five because the signature alone does not fit, three because of this frame's header. We state the five conditions under which that holds and what each relaxation buys (Sec. VII).
- **A frame as one object, and which constraint binds.** The design and its baseline are frames that were emitted, decoded and verified, and their sizes are measured lengths (Sec. IV); the lean format is 42% smaller than the one we built first. Freshness, not the MTU, caps the batch at four records. A delta-coded frame that depends on its predecessor misses the verifiability target at its own loss budget, so every frame carries one self-contained record. Counting cryptography alone, one CPU core serves a

neighbourhood of 296 nodes with Ed25519 and of 10 with BLS; the interpreted receiver we built serves 57 on one core and needs three of the board’s four for the design’s neighbourhood.

In short, batching and header design decide what can run, and among standard 64 B elliptic-curve signatures the choice of scheme does not. AUTHBC introduces no new cryptographic primitive. What is measured, what is simulated and what is only modelled is stated with each result and collected in Sec. X.

II. BACKGROUND AND RELATED WORK

Per-message authentication. MAVLink 2 signing appends a link identifier, a 48-bit timestamp and a 48-bit truncated SHA-256 tag computed with a shared secret [1]. It authenticates a *link*: any holder of the key can produce any message, so a record cannot be attributed to one vehicle afterwards. Rotta and Mykytyn likewise attach an HMAC and a timestamp to each broadcast telemetry message of a swarm [3]. Vehicular broadcast uses public-key signatures: ECDSA over each message, with the signer’s certificate in some messages and an eight-octet digest of it in the others [2], [4]. That is the trust model we adopt, and the per-message cost we start from.

Signing streams. To amortise one signature over many packets, Gennaro and Rohatgi sign the first packet and chain the rest by hashes [5]; one lost packet breaks the chain. Wong and Lam sign a hash tree over a block and send in every packet the hashes needed to reach the root [6]: each packet verifies alone, and the signing cost is amortised but the bytes are not. EMSS tolerates loss by carrying several hashes per packet and a periodic signature packet, and TESLA replaces the signature by a MAC whose key is disclosed later [7], [8]; TBRD applies TESLA to UAS Remote ID and reports about half the overhead of signatures [9]. Erasure codes spread a block’s hashes and signature over its packets so that the block verifies when enough of them arrive, for 10 B per packet in blocks of sixteen at 5% loss and a latency of one block [10]. These schemes spread *one authenticator across packets*, so packets depend on each other or on a later one. We put *several records in one packet*, which a small telemetry record allows: each frame stays self-verifying and the cost moves to latency, at the sender. TESLA delays verification at the receiver instead, and a disclosed MAC key gives no non-repudiation, which a ledger needs. Section IX places these schemes at our operating point.

Aggregate and batch verification. BLS signatures aggregate [11], [12], and certificateless aggregate schemes (CLAS) for vehicular networks build on such constructions [13]. Zhang *et al.* verify many signatures at once at a roadside unit [14], and Rajasekaran *et al.* batch-authenticate a group of drones in a pairing-based scheme, reporting a communication cost that grows linearly with the group [15]. All of these reduce the verifier’s *work*, and an aggregator may forward one short aggregate in place of many signatures; but each message, as its signer sends it, still carries its own signature material (Sec. IX).

Per-record signing: one frame per record, 145.8 B each



AUTHBC: one frame per four records, 173.0 B

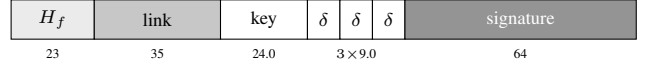


Fig. 1. The lean frame, to scale (bytes below). Per-record signing repeats header, chain link and signature with every record; the design carries them once per four records, and the three records after the first are coded as differences (δ).

Hash-linked records. Linking each record to the hash of its predecessor to make a sequence tamper-evident is due to Haber and Stornetta [16]; blockchain uses on UAVs are surveyed in [17]. We take the chain as given and ask what it costs on air.

Small payloads. That a signature leaves little room in a small frame is known: Gündoğan *et al.* find 9 B of application data left beside an Ed25519 signature in an 802.15.4 frame [18], and post-quantum signatures do not fit one 5G broadcast block [19]. For Remote ID, a 64-octet signature is pagged over 25-octet Bluetooth frames with a parity page [20]: the spreading and recovery that our scope excludes. ADS-B is the extreme case: its 112-bit messages leave 56 bits for data, and proposals for public-key authentication distribute one signature over several messages that receivers buffer [21]. Compact encodings [22] and delta coding of time series [23] are established; a lossy link limits how far a delta chain may run (Sec. VIII-B).

Channel models. Broadcast frames are not acknowledged and their contention window never doubles, so Bianchi’s saturation model of the 802.11 DCF [24] does not apply as it stands; we use the broadcast model of Ma and Chen [25], [26] and validate it against ns-3 [27]. For LoRa, Haxhibeqiri *et al.* model single-gateway scalability [28] and Zirak *et al.* report air-to-air delivery against range on hardware [29].

III. SYSTEM AND THREAT MODEL

Network and traffic. N UAVs share one channel and all hear one another: one collision domain, which we call a neighbourhood of N nodes. Each produces Λ telemetry records per second and broadcasts them; there are no acknowledgements and no retransmissions. A record holds a millisecond timestamp, latitude, longitude and altitude, three velocity components, battery level and flight mode, all integers. Our primary operating point is $\Lambda = 50$ records/s with a freshness bound $D_{\max} = 100$ ms: the rate is PX4’s companion-computer stream rate [30] and the pair meets the direct UAV-to-UAV broadcast requirements of 3GPP [31]. A relaxed point, 20 records/s and 250 ms, is reported for comparison.

Ledger. Each UAV keeps an append-only chain of its own records, $prev_hash_i = \text{SHA-256}(rec_{i-1})$, and stores the verified records it hears from others. There is no consensus layer; the subject of this paper is authenticated dissemination.

TABLE I
BYTE BUDGET OF A FRAME BY FIELD: FIVE-DIGIT SENDER IDENTIFIER,
ONE HOUR INTO A 50 HZ FLIGHT (SENDER 0, FIRST FRAME OF A FLIGHT,
IN PARENTHESES).

field	content	first	lean
(map)	CBOR map header	1	1
v, t	version; frame type	6	4
src	sender identifier, 16 bit	7 (5)	4 (2)
base_seq	sequence no. of first record	14 (10)	6 (2)
n	records in the frame	3	2
recs	key, length of records	6	3
auth	key, length of signature	7	3
header H_f		44 (38)	23 (17)
chain link		32 per record	35 per frame
signature g_a (Ed25519, ECDSA)		64	64
keyframe record s_k		59.9*	24.0
delta record s_δ , 50 ms apart		44.0*	9.0

* includes that record's own 32 B chain link.

Adversary. The adversary controls the radio: it reads, injects, replays, modifies, delays and drops frames, and may operate many radios. It holds no honest UAV's private key.

What is protected. (i) *Origin and integrity*: a stored record was produced by the UAV it names and was not altered, by a public-key signature checked before anything is stored. (ii) *Replay*: sequence numbers are strictly increasing per sender; a repeated frame is refused. (iii) *Equivocation*: two validly signed records for one sequence number are detected, and the two frames that carried them are kept. The signature covers a frame, so a record is evidence only with its frame; a third party checks the pair with the sender's public key alone. (iv) *Continuity*: omission or reordering within what a receiver holds is detected through the chain.

What is not. Telemetry is sent in clear. Jamming is not resisted: loss appears as the robustness measure V below, not as a security property. Nor is flooding: every injected frame that names a known sender costs a receiver one decode and one verification before it is refused. Replay is refused by sequence number, not by time, so a frame that an attacker withholds and delivers later is accepted, authentic and in order, however stale; refusing it needs a clock shared with the sender, which we do not assume. A UAV with a valid key can sign false data; only inconsistent falsehood is detectable. Key distribution, revocation and admission are assumed: receivers hold senders' public keys, and we charge the bytes of delivering certificates separately (Sec. VIII-E). Senders are not anonymous.

Verifiability. V is the probability that a record a UAV generates is received, decoded and verified by a given neighbour. We require $V \geq 1 - \epsilon$ and take $\epsilon = 0.05$ throughout.

IV. THE AUTHBC FRAME

A frame is one canonical CBOR map [22] with integer keys (Table I). `src` and `base_seq` name the sender and the sequence number of the first record; `link` is the hash of the record before it; `recs` is the record stream and `auth` one signature. A record in the stream is nine variable-length integers. The first record of a frame is a *keyframe*, carrying

absolute values (24.0 B on average); each later one is a *delta* from the record before it (9.0 B, the floor of the format, at the 20 and 50 ms spacings of our two operating points). Sender and sequence number are not repeated in the records.

What is signed is not the stream but the canonical ledger records, each with its own `prev_hash`. The receiver rebuilds them — the first `prev_hash` is `link`, each later one is the hash of the record just rebuilt — and only then checks the signature. A frame that cannot be decoded therefore cannot be verified, and nothing unverified is stored. Each verified record is offered to the ledger store, which refuses a sequence number that is not newer, keeps an equivocating pair as evidence, and flags a link that contradicts a record it holds. The state against which a later frame would be decoded advances only after a frame has been verified and stored, so a replayed frame cannot desynchronise a stream. Flipping any single bit of a frame makes verification fail (all 1392 positions of a four-record frame were tried).

When a frame is lost, the next one still decodes and verifies, because it starts with a keyframe. Its `link` then names a record this receiver does not hold: the gap shows in the sequence numbers, and the link is stored unchecked, to be checked by any party that holds the missing record. A late copy of the lost frame is refused, since sequence numbers must increase; the gap is not repaired.

One link per frame. The ledger still chains every record. Only the first link of a frame travels; the receiver recomputes the others. Inside a frame the order and content of records therefore rest on the frame signature alone, where in the first format below every record also carries its own hash on air. A frame is signed and accepted as a unit, so those hashes were redundant and a receiver loses no check; but one mechanism remains where there were two.

The first format. For comparison we keep the format we built first: text keys, and in every record its own chain link, sender identifier and sequence number. Its header is 44 B, of which 29 B are key names. That format is implemented and frozen. Its sizes in this paper are sums of separately measured parts, and its delta-coded frames were never emitted; we mark such sizes wherever they appear.

Design parameters. A configuration is the batch size b ; whether a signature covers each record (*inline*; `auth` is then an array of b signatures) or the frame (*batch*); the signature scheme (Ed25519 [32] and ECDSA P-256 [33] at 64 B, BLS12-381 [11] at 96 B in the library used); and the reference interval r : one frame in r starts with a keyframe, the others with a delta against the previous frame.

V. CONSTRAINTS

Bytes. With header H_f , link field L , signature g_a , keyframe s_k and delta s_δ , a self-contained batch frame costs, per record,

$$B(b) = \frac{H_f + L + g_a + s_k + (b-1)s_\delta}{b}. \quad (1)$$

$B(1)$ is our baseline: one record, one signature, one frame. Signing each record of a shared frame replaces g_a by $b g_a$.

Equation (1) is exact to within a byte of CBOR length prefix; for the lean design and its baseline we report the lengths of emitted frames.

Freshness. The oldest record in a frame waits for the frame to fill. Adding the channel-access and airtime delay T_{ch} and the verification time t_v ,

$$D(b) = \frac{b}{\Lambda} + T_{ch} + t_v \leq D_{\max}. \quad (2)$$

Fill time dominates. At both operating points $\Lambda D_{\max} = 5$: five records fill in exactly $D_{\max}$ and leave nothing for the channel, so $b \leq 4$ whatever the encoding or the scheme. Equation (2) counts the whole filling period b/Λ . A record may describe a state up to one sampling interval old; and with b/Λ the bound is on the *peak age* of information [34] at a receiver that loses no frame: the delay of a frame's newest record plus the interval to the next frame. Counting only the wait $(b-1)/\Lambda$ would admit $b=5$ and 36.4 B per record. The 1500 B MTU would admit 31 records in the first format and 151 in the lean one.

Verifiability. If one frame in r is self-contained, frame j of a group verifies only if it and the $j-1$ frames before it arrive. With independent frame loss p ,

$$V(r, p) = \frac{1}{r} \sum_{j=1}^r (1-p)^j, \quad (3)$$

which equals $1-p$ at $r=1$ and is smaller for every $r>1$. A target $V \geq 1-\epsilon$ with $\epsilon \leq p$ therefore admits $r=1$ only. The same argument forbids splitting a signed unit over n frames, which verifies with probability $(1-p)^n$.

Channel. For a configuration — a frame size and a frame rate per node — $N_{\max}$ is the largest neighbourhood in which the mean delivered fraction of frames is at least $1-\epsilon$. We find it by simulation (Sec. VI). As a conservative companion we give N_{sat} , the largest N whose offered frame rate does not exceed the saturation throughput of the broadcast DCF model [26].

Receiver CPU. A receiver verifies every neighbour's frames. With k signatures per frame, verification time t_v and hash time t_h , one neighbour costs $(\Lambda/b)k t_v + \Lambda t_h$ seconds of CPU per second, and one core serves

$$N_{\text{cpu}} = 1 + \left\lfloor \frac{1}{(\Lambda/b)k t_v + \Lambda t_h} \right\rfloor \quad (4)$$

nodes, itself included.

Payload. A frame that verifies alone must fit the link's maximum payload M :

$$M \geq H_f + L + g_a + s_k. \quad (5)$$

VI. METHOD

Implementation. Formats, sender, receiver and store are one Python code base with four record encodings (JSON, CBOR, MessagePack, delta) and three signature schemes behind common interfaces, each scheme checked against published test vectors (for Ed25519, those of [35]).

Telemetry. Records come from a seeded random-walk generator: 30 seeds of 1000 records at 50 ms spacing, that of the

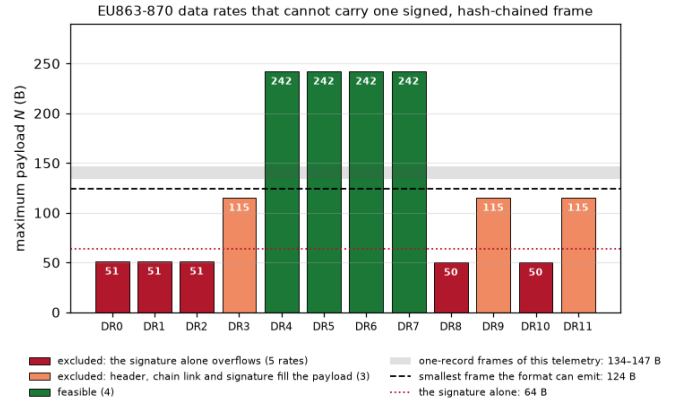


Fig. 2. Payload limits of the twelve EU863-870 data rates against the smallest lean frame and the frames of this telemetry (64 B signature).

relaxed point. Header fields are sized for a five-digit sender identifier one hour into a 50 Hz flight. Section VIII-G checks the generator against flight logs.

Timing and energy. Signing, verification, hashing and encoding are timed on a Raspberry Pi 4 (medians; Ed25519 sign/verify 88/259 μs , ECDSA 136/327 μs , BLS 2.73/8.61 ms, SHA-256 2.75 μs). Sender energy is metered with an INA219 on the supply rail, five repetitions per configuration, idle power subtracted: the first format on that board, the lean format on a second board and sensor of the same types.

Channel simulation. ns-3.48 [27]: IEEE 802.11a at 6 Mb/s, ad hoc mode, broadcast frames, all nodes in range at equal received power, so there is no capture and no hidden terminal. Each node sends one frame every b/Λ seconds, at an instant drawn uniformly within each period, so the relative phases of the nodes change from frame to frame. We do not use a strictly periodic source, because it freezes every phase for a whole run: a run then samples one phase configuration, in which a few nodes may collide in every period while the others lose nothing. The two sources deliver the same on average — over 14 points and 1260 runs the means differ by 0.0004 ± 0.0009 — but the strictly periodic one varies 6–23 times more from run to run (Sec. X).

A run lasts 20 s and each point is 30 runs. $N_{\max}$ is the largest simulated N whose mean delivered fraction is at least 0.95, every smaller simulated N also passing. Its interval is the 2.5–97.5 percentile of that quantity over 10 000 bootstrap resamples of the seeds. The saturation models were validated separately against saturated ns-3 runs (Sec. VIII-H).

VII. LOW-RATE LINKS: WHERE NO BATCH HELPS

LoRaWAN limits the payload of a frame by data rate, and its EU863-870 plan defines twelve data rates [36]: seven LoRa (DR0–DR6), one FSK (DR7) and four LR-FHSS (DR8–DR11). By Eq. (5) a frame that verifies alone needs a header, a chain link, a signature and one record that decodes alone. The smallest such frame the lean format can emit at all — sender 0, sequence 0, every field zero — is $16+35+64+9 = 124$ B, and

TABLE II

DOES ONE SELF-CONTAINED LEAN FRAME FIT? FLOOR: EVERY FIELD ZERO. RANGE: FRAMES OF THIS TELEMETRY.

	floor (B)	range (B)	50/51 B 5 rates	115 B 3 rates	222/242 B 4 rates
64 B signature, chain link	124	134–147	no	no	yes
... link not sent on air	89	99–112	no	yes	yes
48 B BLS signature	108	118–131	no	no*	yes
13 B symmetric tag	72	82–95	no	yes	yes

* a frame of zeros would fit; no frame of this telemetry does.

we emitted, signed and verified it. Header, link and signature alone are 115 B.

Figure 2 sets that against the payload limits. Five data rates (DR0–DR2, DR8, DR10) allow 50 or 51 B: the signature alone does not fit. Three (DR3, DR9, DR11) allow 115 B: header, link and signature fill the payload and no record fits. Four (DR4–DR7) allow 222 or 242 B and are feasible. **Eight of twelve data rates cannot carry one signed, chained frame**, in either format and under either of the standard’s payload tables. None of the eight depends on the telemetry, since each is excluded against the format’s floor; three of them depend on the format (condition v).

Scope. The result holds under five conditions. (i) The limits are LoRaWAN’s regional parameters; the LoRa PHY itself carries 255 B [37] and is not excluded. (ii) Every frame carries a public-key signature. (iii) There is no loss recovery. (iv) $\epsilon \leq p$, which by Eq. (3) forbids spreading a signature over several frames; at $p = 0.01$ and $\epsilon = 0.05$ a signed unit may span five frames and the exclusion no longer holds as stated. (v) The frame is the format of Sec. IV. The five rates of 50 and 51 B are excluded by the signature alone, whatever the format; at 115 B a signature and a 32 B hash leave 19 B, which this format’s header and framing fill exactly. A hand-packed layout would leave part of them for a record, less than the 24.0 B our keyframe averages.

What each relaxation buys (Table II). Taking the chain link off the air frees the three 115 B rates: they are excluded by the link, not by the signature. The smallest standardised signature, 48 B BLS12-381 [38], fits a 51 B payload and leaves 3 B; at 115 B a frame of zeros would fit (108 B) but the smallest frame of our telemetry is 118 B, a three-byte miss that we do not count as an exclusion. A 13 B symmetric tag, as in MAVLink 2, fits the 115 B rates, but at 50 or 51 B header and link already take 51–58 B: a hash-chained frame does not fit there under any authenticator.

What the feasible rates carry. At DR5 under the 1% duty cycle a 242 B frame holds 8 records and sustains 0.20 records/s per node (5 records and 0.13 in the first format). The batch must be sized at the record spacing it implies: a delta that spans 5.5 s costs 13.6 B, not the 9.0 B it costs at 50 ms. The ns-3 LoRaWAN module carries at most 222 B, where a frame holds six records. Simulated with a 218 B payload every 36.4 s (0.165 records/s per node), on one channel with one demodulator and no capture, the largest neighbourhood with mean $V \geq 0.95$ is three nodes (30 seeds, 95% interval

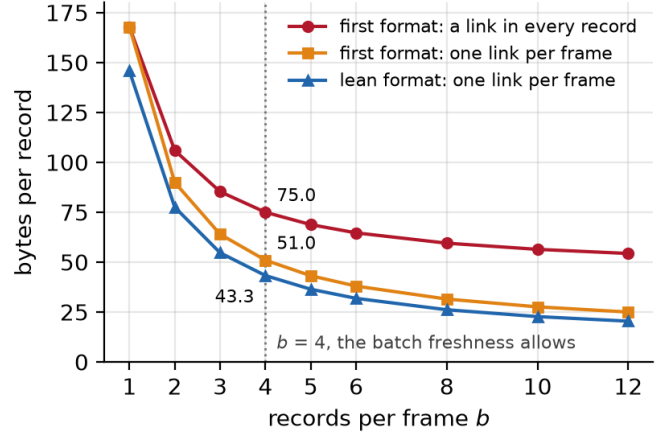


Fig. 3. Bytes per record against batch size, by where the chain link sits (delta records throughout). First-format curves are sums of measured parts; lean frames are emitted.

[2, 3]; one if 95% of runs must each reach it); the closed-form model of Haxhibeqiri *et al.* [28], fitted to hardware interference measurements and evaluated at the same operating point, gives four. Such a neighbourhood carries about 0.5 records/s in all, where the 802.11 design of Section VIII carries some 6 200. On these links the ledger records provenance every few seconds; it is not the 802.11 design at a smaller scale.

VIII. RESULTS ON 802.11

A. The design ladder

Table III takes the design apart. A baseline that signs every record and sends it alone confounds two things, so rows 2 and 6 separate them: four records share a frame but each keeps its signature. In the lean format that step takes 145.77 B per record to 93.00: the header and the chain link are paid once per four records, and the three records that follow the first can be coded as deltas. Sharing the signature as well gives **43.25 B, 70.3% below the baseline**; row 7 is the same frame without delta coding, 54.50 B. In the first format the steps are sharing the frame (174.25 to 141.25 B), sharing the signature (93.25) and delta coding (74.96, 57.0% below its baseline). It ends higher because each of its records carries a 32 B link of its own, which no batch size amortises.

Figure 3 shows where the link belongs. Moving the first format’s link from the record to the frame saves $32(1 - 1/b)$ B per record: 24 B at $b = 4$, which gives 51.0 B. The lean header and record coding account for the remaining 7.7 B. With a link in every record, 12 records per frame still cost 54.3 B each, more than one link per frame costs at four.

An exhaustive search over placement, batch size 1–12 and reference interval returns row 8 as the byte-minimal configuration that meets the freshness, verifiability and MTU constraints, and row 4 for the first format.

The choices are made together because they are not independent. One signature per frame saves $g_a(1 - 1/b)$ per record, and delta coding saves $(1 - 1/b)$ of the difference between a

TABLE III

THE DESIGN LADDER AT 50 RECORDS/S AND $D_{\max} = 100$ MS. N_{sat} : SATURATION BOUND. $N_{\max}$: LARGEST NEIGHBOURHOOD WITH MEAN DELIVERY ≥ 0.95 IN NS-3 [95% INTERVAL]. CPU: SHARE OF ONE RECEIVER CORE SPENT VERIFYING AND HASHING AT $N_{\max}$.

				with cert., one per		N_{sat}	N_{max}	CPU
configuration	b	frame (B)	B/record	5 frames	500 ms			
<i>First format: text keys, a chain link in every record[†]</i>								
1 sign every record, one record per frame (baseline)	1	174.3	174.25	213.05	188.41	18	32 [32, 32]	41%
2 sign every record, share the frame	4	565.0	141.25	150.95	149.41	24	56 [56, 56]	72%
3 one signature per frame	4	373.0	93.25	102.95	101.41	30	76 [75, 76]	25%
4 ... and delta records	4	299.9	74.96	84.66	83.12	34	88 [88, 89]	29%
<i>Lean format: integer keys, one chain link per frame</i>								
5 sign every record, one record per frame (baseline)	1	145.8	145.77	184.57	159.93	20	35 [35, 35]	45%
6 sign every record, share the frame (delta records)	4	372.0	93.00	102.70	101.16	30	76 [76, 77]	98%
7 one signature per frame, no delta coding [†]	4	218.0	54.50	64.20	62.66	42	108 [107, 109]	36%
8 one signature per frame, delta records (the design)	4	173.0	43.25	52.95	51.41	52	124 [124, 125]	42%

[†] size is a sum of measured parts; rows 5, 6 and 8 are emitted frames.

TABLE IV

VERIFIABILITY AT FRAME LOSS $p = 0.05$ WITH ONE SELF-CONTAINED FRAME IN r ($b = 4$, LEAN FORMAT; 30 SEEDS, 95% INTERVAL).

r	B/record	V measured	Eq. (3)	$V \geq 0.95$
1	43.25	0.952 [0.947, 0.957]	0.950	yes
2	41.38	0.921 [0.916, 0.927]	0.926	no
4	40.44	0.876 [0.863, 0.889]	0.881	no
16	39.73	0.655 [0.618, 0.690]	0.665	no

self-contained record and a delta, since each frame must start with a self-contained one. Both vanish at $b = 1$, where rows 1 and 5 already are the design. Batching is the choice the other two gains depend on, and the deadline sets it.

Among the 64 B schemes the signature scheme cannot move the bytes or the capacities of these rows: Ed25519 and ECDSA give frames of one size. They differ in verification time (259 against 327 μ s), which enters Eq. (4) only. BLS adds 32 B per frame and, as shown below, fails on CPU.

B. Loss, and why every frame carries a keyframe

Delta coding is usually run with a keyframe every K records. With $K = 16$ and $b = 4$ only one frame in four decodes without its predecessor, and the frames that follow a lost one cannot be decoded until the next keyframe. Table IV measures this with the codec in the loop — 7200 frames per cell through the sender, an independent loss process and the receiver: at the loss rate the system is specified for, $r = 4$ verifies 0.876 of records, not 0.95. Equation (3) predicts the measurement at every r . Under bursty loss a dependent design loses less — a Gilbert process with the same mean and bursts of four frames gives 0.929 at $r = 4$ — but it does not reach $1 - p$. A cleaner link would admit a longer interval ($r = 4$ at $p = 0.02$, $r = 8$ at $p = 0.01$), and the most a longer interval can save is 3.5 B per record. The loss rate here is a specification, not a measurement: at the 2.3×10^{-4} we measured between two radios (Sec. VIII-H) every interval in the table meets the target, $r = 16$ with $V = 0.998$. The design is sized for the budget and therefore uses $r = 1$: each frame

TABLE V

DELAY OF THE OLDEST RECORD IN A FRAME ($b = 4$, 50 RECORDS/S); CHANNEL DELAY SIMULATED AT THE HEaviest LOAD THAT STILL DELIVERS 95% OF FRAMES.

	fill (ms)	channel (ms)	verify (ms)	total (ms)	margin (ms)
Ed25519, mean channel delay	80	0.76	0.26	81.0	19.0
Ed25519, 99th percentile	80	2.40	0.26	82.7	17.3
Ed25519, worst observed	80	3.49	0.26	83.8	16.2
BLS, worst observed	80	3.49	8.61	92.1	7.9

starts with a keyframe, costs 43.25 B per record, and loses only its own records when it is lost.

These figures give every frame the same loss probability. At a fixed bit error rate a longer frame is lost more often, and a batch is a longer frame. On a channel that loses 5% of the baseline's one-record frames, the design's frame is lost with probability 0.057: 0.943 of its records are expected to verify, below the target, and the measurement, 0.947 [0.941, 0.952], agrees with that expectation. The design reaches 0.95 only while one-record frames are lost at most 0.044 of the time. The capacities of Table III are simulated at equal, strong received power, where frames are lost by collision and not by bit error, and carry that qualification.

C. Freshness

Table V adds the terms of Eq. (2). The frame fills in 80 ms; channel access and airtime add 0.76 ms on average and 3.49 ms at worst; one Ed25519 verification adds 0.26 ms. The oldest record is verified within 83.8 ms, leaving 16.2 ms of the 100 ms budget. Batching b records needs a record rate above $b/D_{\max}$, here 40 records/s: at 20 records/s with the same 100 ms bound the batch is one record and nothing is gained.

D. Capacity

Bytes matter because they are airtime. Figure 4 shows delivery falling as nodes are added, and Table III gives the resulting $N_{\max}$ for every row. The lean baseline holds 35 nodes at 95% delivery and the design 124 (3.5 $\times$); in the first format

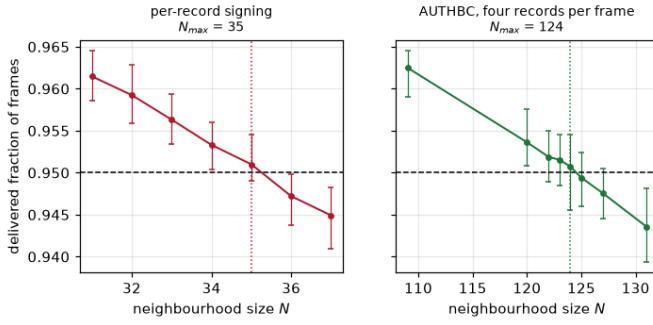


Fig. 4. Delivered fraction of frames against neighbourhood size in ns-3: lean baseline (row 5) and design (row 8), mean and range of 30 runs.

TABLE VI

$N_{\max}$ OF BASELINE AND DESIGN AT BOTH OPERATING POINTS (NS-3, 30 SEEDS, 95% INTERVAL).

	baseline	design	ratio
<i>50 records/s, 100ms</i>			
first format	32 [32, 32]	88 [88, 89]	2.8×
lean format	35 [35, 35]	124 [124, 125]	3.5×
<i>20 records/s, 250ms</i>			
first format	78 [77, 78]	219 [215, 221]	2.8×
lean format	85 [85, 86]	306 [306, 308]	3.6×

the figures are 32 and 88 ($2.8\times$). The larger step is sending a quarter as many frames: row 6, which shares the frame but not the signature, already reaches 76 ($2.2\times$ the baseline), and sharing the signature as well adds $1.6\times$. Each frame pays a preamble, an inter-frame space, a backoff and MAC framing whatever its length — about $173\ \mu\text{s}$, 47% of what a baseline frame costs the channel — so removing three frames in four is worth more than their bytes. This is the principle of frame aggregation in 802.11n, whose aggregates are addressed to one receiver and exclude broadcast [39]; batching at the application gives a broadcast sender the same effect.

These figures hold the mean over runs to the 0.95 target. Requiring that 95% of individual runs reach it gives 34 and 120 (31 and 86 in the first format), at most four nodes fewer.

Table VI repeats the comparison at the relaxed operating point. The ratio hardly moves — $3.5\times$ and $3.6\times$ in the lean format, $2.8\times$ and $2.8\times$ in the first — and depends on the format, so we quote it with the format.

The capacity envelope: a rule for the crossing. At the 5% level, loss follows a line in airtime. With $f = \Lambda/b$ the frames per second a node sends and T the time one frame holds the medium (its airtime plus DIFS),

$$N_{\max} \approx \frac{0.05}{f(aT + c)}, \quad a = 0.071, \quad c = 8\ \mu\text{s}, \quad (6)$$

where c is twice the simulator's preamble-detection time and a is the one fitted constant. It was fitted to six rows of Table III (1, 2, 4, 5, 6 and 8), which the rule then reproduces within 3.8%. Twelve further configurations were predicted from the rule before they were simulated: seven others of our own, all

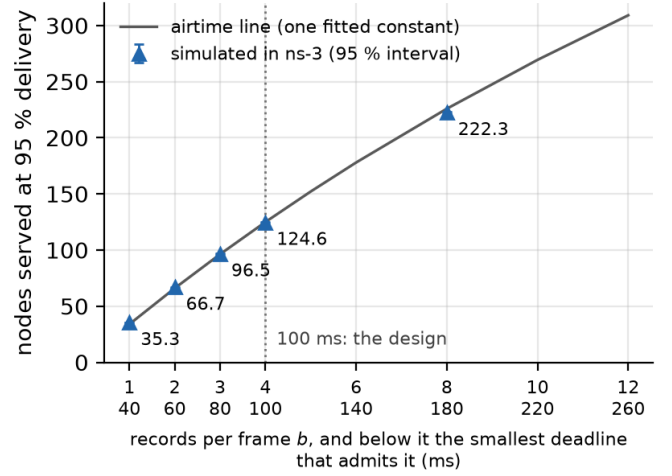


Fig. 5. Nodes the lean design serves at 50 records/s, by batch size and the deadline that admits it: simulated crossings of 0.95 delivery, and the rule.

within 3.4% of it, and the five one-record frames of Table IX, within 4.2% and all above it. A single utilisation ceiling, measured at one point and applied to all, misses the first thirteen (six fitted, seven predicted) by up to 15%, in both directions. Equation (6) is a fit at one loss level and one PHY rate, checked for frames of 94–565 B, 5–50.5 frames/s and 28–306 nodes; the tables quote simulated values, not the rule. All of those configurations hold one record or four. Batches of two, three and eight, which deadlines of 60, 80 and 180 ms would admit, were predicted before they were simulated: delivery crosses 0.95 at 66.7, 96.5 and 222.3 nodes, within 1.6% of the rule and 0.8% of the access-rule model described next. Figure 5 sets them on the rule: it is what a deadline buys at this record rate, and at 40 ms it buys nothing.

Why the crossing is there. The slope is not a free constant. A frame is lost in this scenario in two ways. In the first, two stations that found the medium busy draw the same backoff slot. With occupancy $\rho \approx NfT$ and a contention window of $W = 16$ slots, counting the stations that wait together gives a loss of about $\rho^2/(W(1-\rho))$, hence $a = \rho/(W(1-\rho))$, which is 0.071 at $\rho = 0.53$, the mean occupancy at which loss reaches 5%. This mechanism is known from analyses of vehicular safety broadcast [40], [41]. In the second, which those analyses do not contain, a station decides to send within the $4\ \mu\text{s}$ in which it cannot yet sense another: that is c . Without the first-order step, the count is $\rho[1 - (1 - 1/W)^{\rho/(1-\rho)}]$; with the detection term and nothing fitted, solving it for 5% gives the 18 crossings within 5.5% (mean 2.3%).

An event simulation of the access rule alone, with nothing fitted and none of the simulator's code, reproduces all 18 simulated crossings within 1.2%. For crossings that had not been simulated — the contention window doubled, and loss levels of 2% and 10% — it predicted all six within 1.7%. Doubling the window moves the baseline's crossing only from 35.3 to 39.2 nodes (predicted 39.1): stations that count down for longer wait together more often. The capacities therefore

follow from the access rule and one detection time, and they are the capacities of radios that keep that rule (Sec. VIII-H).

Nodes spread out, and capture. All of this puts every node at one point: equal received power, so a frame that overlaps another is lost everywhere. With the nodes placed uniformly in a disc and power falling as in free space, a receiver keeps the strongest of the frames that start together if it is 4 dB above the rest. The access-rule model with that one rule, read in the simulator’s source, gave 39.5 and 139.7 nodes for baseline and design before the runs. In a disc of radius 15 m, where every station still senses every other by energy, ns-3 gives 39.6 and 140.4; at 100 m, a swarm’s scale, 39.4 and 139.3, at most 0.7% lower. The equal-power capacities of Table III are therefore conservative for unequal power, by 12–13%, and the ratio of design to baseline stays $3.5\times$. Both discs keep every pair in range: hidden stations, which would lower the capacity, are not in this test.

Receiver CPU. Equation (4) charges cryptography only. By it, one core of the Raspberry Pi 4 serves a neighbourhood of 296 nodes when each frame carries one Ed25519 signature and of 77 when every record is signed. At its $N_{\max}$ the design uses 42% of a core, so by this count the channel binds first. Row 6 shows that the count can bind: with a signature on every record it reaches the channel’s limit at 98% of a core. With BLS a verification takes 8.61 ms and one core serves 10 nodes: on this hardware BLS is excluded by CPU well before the channel is reached, in addition to its larger signature. Ed25519 was designed for batch verification [32]: on the Pi 4 a batch of 64 takes 0.46 of the time per signature. The prototype verifies one signature at a time, and ECDSA does not admit batches.

The receiver we built does more than verify, and for it the order is reversed. Decoding a frame and rebuilding its records, in an interpreted prototype, bring the time to receive a four-record frame on the Pi 4 to 1.43 ms, 5.5 times its verification. By that time one core serves 57 nodes: the processor binds before the channel. Run as one receiver process per core, one to four at once, the board serves 56, 106, 150 and 187 nodes. Four cores give 3.36 times one, less than we had written down beforehand, so 124 nodes need three of them. The figures of Eq. (4) are ceilings for a compiled receiver.

Other PHY rates. We simulated one rate. For the others we give only the saturation bound, which at 6 Mb/s rises from 20 to 52 nodes. With the same model at 12 and 24 Mb/s, and on the 10 MHz channel of 802.11p at 3–12 Mb/s, the design-to-baseline ratio of N_{sat} runs from $1.7\times$ to $4.3\times$ over both formats, growing with the rate: a faster PHY shortens the payload but not the fixed cost of a frame.

E. Certificates

The rows so far assume that receivers already hold the senders’ public keys. Vehicular practice sends a full certificate in every fifth message and an eight-octet digest in the others [4]. Charging that to every frame — a 162 B certificate, so 38.8 B per frame on average — gives the first “with cert.” column of Table III: 52.95 B per record for the design against 184.57 for the baseline. That reading favours the design: the

TABLE VII
SENDER ENERGY PER RECORD ON A RASPBERRY PI 4: METERED (MEDIAN AND RANGE OF FIVE REPETITIONS) AGAINST TIME $\times$ POWER. LEAN ROWS: SECOND BOARD AND SENSOR.

	metered (μJ)	model (μJ)	gap
first format, every record signed	118.8 [118.5, 119.9]	105.8	12.3%
first format, one signature per 4	58.4 [58.1, 59.3]	54.3	7.5%
lean format, every record signed	273.1 [270.6, 276.9]	284	−3.9%
lean format, one signature per 4	155.4 [155.1, 155.8]	163	−4.4%

practice it copies sends ten messages a second, so one in five is one certificate in 500 ms, and a baseline of 50 frames/s charged per frame sends five times that. The second column charges one certificate per 500 ms at any frame rate, 51.41 B against 159.93; batching then amortises only the digest. These are explicit certificates; implicit ones are shorter (a signed message of at most 226 B against 330 B [4]) and are not priced, so both columns are upper bounds.

F. Energy

Table VII meters the sender in both formats. In the first, encoding, hashing and signing a record costs 58.4 μJ batched against 118.8 μJ when every record is signed, 51% less. The model composes separately measured times with one power, the median over these same metered runs, so the comparison tests the times: the model is 7.5–12.3% low, the gap being frame assembly in an interpreted prototype.

The lean sender, whose model is its timed frame (0.87 ms for four records) at that power, meters 155.4 μJ per record against 273.1 unbatched, 43% less, and reads −4.4% and −3.9% against its model. Its rows were metered on a second board and sensor, where a control repeating the first row gives 116.2 μJ . Between the formats the order is the reverse of the bytes: the lean sender costs 2.7 times the energy per record of the first format’s. Its codec was written for the size of the frame and is, in this prototype, the slower to run, so the design that is smallest on air is not the cheapest to compute.

Receiver and radio energy are modelled, not metered, and for the first format only, the radio’s power coming from a unicast measurement on another band: adding one receiver and the radio, the model gives 131.7 against 376.8 μJ per record.

G. Is the synthetic telemetry representative?

Every size above comes from one generator, whose delta record is exactly 9.0 B in every sample — the floor of the format. We ran twelve public PX4 flight logs (four quadrotors, two hexa/octorotors, three fixed wings, three VTOLs; 29 092 in-flight records), chosen by a rule fixed before any was opened, through the same encoder. The logs carry position at 5 Hz, so the comparison is at 0.2 s spacing and above, not at the 50 Hz of the operating point. At 0.2 s a real delta record averages 11.1 B against the generator’s 10.8 B, and a real keyframe 22.0 B against 24.0. The spread is by speed: the six multicopters, mostly hovering, average 10.0–10.5 B and the three fixed wings up to 13.2 B. With real sizes at 0.2 s the design saves 69.1% (68.2% in the least favourable log) where

TABLE VIII

BYTES PER MESSAGE OF CERTIFICATELESS AGGREGATE SCHEMES AS REPORTED IN [13] (WITH A 67 B MESSAGE), AND PER RECORD OF AUTHBC WITH CERTIFICATES CHARGED PER FRAME.

	bytes per message
Wang <i>et al.</i> *	859
Liang <i>et al.</i> *	735
Xu <i>et al.</i> *	596
Cahyadi <i>et al.</i> *	583
Li <i>et al.</i> [13]	583
AUTHBC baseline (row 5), with certificates	184.57
AUTHBC design (row 8), with certificates	52.95

* total per message as reported in [13].

the generator gives 69.4%. The generator is representative to within about one percentage point, and optimistic for fast vehicles.

For the operating rate itself we flew PX4 in software-in-the-loop — its own estimator, logger and MAVLink module on a simulated quadrotor — with position logged every 20 ms. Over 14 317 in-flight records, at up to 10 m/s, every delta record is 9.0 B, the floor of the format; the keyframe averages 20.9 B and the design costs 42.47 B per record, 1.8% less than with the generator’s records. The same autopilot’s 50 Hz position stream holds a 20.0 ms period to within ± 0.4 ms (central 68%).

H. Validation of the models

Models against the simulator. The broadcast model [26] matches saturated ns-3 runs within $\pm 0.51\%$ over $N = 5$ –50 (30 seeds).

The access rule against two radios. On two Raspberry Pi 4 radios, one transmitter alone loses 4 frames in 17 600 ($p = 2.3 \times 10^{-4}$). With both radios saturated, 12.4% of the frames put on air are lost where the access rule gives 11.8%, a prediction registered before the run. Below saturation the agreement is poorer: as the medium fills from half to 85% the radios lose 1.13, 0.79 and 0.56 times what the model predicts, and the registered prediction held at the first two loads and failed at the third. The cause is not established. Below saturation a radio often holds two frames of its own, and a radio of this family sends its queued frames closer together than the standard allows — one per 2.019 ms against 2.078 ms — as commercial cards are known to [42]. That state is common with two busy radios and rare at the operating point. But a follow-up that switched off the vendor’s frame bursting, with four further predictions registered, failed all four: bursting has no demonstrated effect on the loss. No capacity was measured: the capacities in this paper rest on simulation.

IX. COMPARISON WITH PUBLISHED SCHEMES

Stream signing. Table IX places the schemes of Section II at our operating point. None is implemented here: each row is the lean one-record frame with that scheme’s authenticator in place of the signature. All of them send a frame per record — 50.0 frames/s per node (50.5 for EMSS, with its signature

TABLE IX

STREAM-SIGNING SCHEMES AT 50 RECORDS/S: THE LEAN ONE-RECORD FRAME WITH EACH SCHEME’S AUTHENTICATOR, SIZED FROM ITS DEFINITION. $N_{\max}$ SIMULATED AS IN TABLE III.

scheme	auth. B/record	total B/record	N_{sat}	$N_{\max}$
MAVLink 2 tag [†]	13	93.8	23	42 [41, 42]
TESLA ^{†‡}	24	105.8	22	40 [40, 40]
Gennaro–Rohatgi [‡]	32	113.8	21	39 [39, 39]
EMSS [‡]	65	147.3	20	34 [34, 34]
Wong–Lam tree, block of 4	129	210.8	17	28 [28, 29]
signature per record (row 5)	64	145.8	20	35 [35, 35]
AUTHBC design (row 8)	16	43.3	52	124 [124, 125]

[†] no non-repudiation.

[‡] a packet does not verify alone: $N_{\max}$ is an upper bound.

packets) where the design sends 12.5 — so the fixed channel time of a frame is paid for every record whatever the size of the authenticator. A 13 B tag in place of a 64 B signature raises $N_{\max}$ from 35 to 42 nodes and gives up non-repudiation; four records in a frame raise it to 124. Where a packet does not verify alone the simulated figure counts frames delivered, not records verified, and is an upper bound. Tree chaining is the one scheme that keeps what a ledger needs — every packet verifies alone and its origin can be shown to a third party — and it pays 129 B per packet for it. These schemes amortise the cost of signing; on a contended channel the cost that binds is the frame.

Aggregate signatures. The nearest published comparators are certificateless aggregate signature schemes for vehicular broadcast. Table VIII lists what one signer sends per message. For n messages that is n times as much, since each carries its own identity, key and signature components. Aggregation acts afterwards, at a roadside unit: it reduces the verifier’s work and what the unit forwards — in the one scheme read at source, 388 B for a single signature and 784 B for the aggregate of a hundred [43]. Batching acts before, on a sender’s own records. The two act on different links, and the table should not be read as a ranking: those schemes provide conditional privacy, which AUTHBC does not; their group elements are 128 B where our signature is 64 B; and their figures are for a different message. What the comparison does show is that per-message authentication of this kind costs several hundred bytes, and that aggregation alone does not change that.

X. LIMITATIONS

Capacities are simulated. On hardware, contention was measured between two radios only (Sec. VIII-H). The capacities in Table III come from ns-3 with one collision domain, equal received power and static nodes. Unequal power with capture was simulated separately and raises them (Sec. VIII-D); hidden terminals, spatial reuse and mobility are absent. Send instants are redrawn within each period (Sec. VI). PX4’s own stream, in simulation, holds its period to about ± 0.4 ms, which is nearer to the strictly periodic source and its wider spread between runs. Three to five radios are the next

measurement; the model’s predictions for them are registered in the repository.

Telemetry is synthetic. Record sizes come from a generator. The check against flight logs covers twelve logs of one autopilot at 5 Hz position resolution; at 50 Hz it rests on one simulated flight of the same autopilot.

One PHY, generated loss. Capacity was simulated at 6 Mb/s only. Both operating points have $\Delta D_{\max} = 5$ and so a batch of four; batches of two, three and eight were simulated for the capacity rule alone, and nothing else here is evaluated at them. The loss experiment uses independent, Gilbert and length-dependent loss, all generated at the frame level, not a channel trace; under the last, the design is below its target where the baseline’s frame loses 5% (Sec. VIII-B). The stream-signing schemes of Table IX are sized from their definitions and simulated as frames; none is implemented.

Energy and CPU. Only the sender is metered, with a sensor never checked against a reference load; receiver and radio energy are modelled. The lean sender costs more energy per record than the first format’s. The receiver’s CPU share counts verification and hashing, not decoding; counting both, the prototype needs three of the board’s four cores for the design’s neighbourhood, measured without a radio or a dispatcher in the path.

Security scope. Confidentiality, availability, key management and protection against a delaying attacker are outside the design (Sec. III), and the security argument is informal. With one link per frame, continuity inside a frame rests on the frame signature alone, and showing one record to a third party means showing its frame. The receiver therefore keeps every accepted frame, at its on-air length for every b records held.

Post-quantum signatures. An ML-DSA signature is 2420 B [4] and exceeds the 1500 B MTU with any header, so it cannot ride in one frame at any batch size; freshness caps the batch at four, so it could not be amortised either. A post-quantum variant would be a different design, built around fragmentation [4], [44].

Multi-hop. Relayed records and more than one collision domain are not evaluated. A relay cannot re-sign: it carries the originators’ signatures or one BLS aggregate [12]. Both are implemented; neither is measured here.

XI. CONCLUSION

For small telemetry records the frame, not the record, is the unit at which authentication should be costed, and what a frame must carry decides where it can be sent. A frame that verifies alone needs a header, a chain link, a signature and one self-contained record: eight of the twelve EU863-870 LoRaWAN data rates cannot carry that at any batch size, five because the signature alone does not fit and three because of this frame’s own header. Where the frame fits, one signature and one chain link per four records cut the bytes per record by 70.3% and let one 802.11a collision domain carry 124 nodes where per-record signing carries 35. Two constraints shape the design: the batch is capped by freshness, not by the MTU,

and a frame that depends on its predecessor does not meet a verifiability target equal to the loss rate. Batching and header design decide what can run; among 64 B signatures the scheme does not. The capacity figures are simulated; on hardware the access rule behind them held between two saturated radios and not at every load below saturation, and no capacity was measured.

DATA AND CODE AVAILABILITY

Code, result files, simulation scenarios, hardware traces and tests are at <https://github.com/besco1998/fanet-authbc>; every number in this paper is generated from the result files, and a gate re-derives the deterministic ones on each commit.

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DECLARATION OF COMPETING INTERESTS

The author declares no competing interests.

USE OF GENERATIVE AI

An AI coding agent assisted with code, analysis and drafting under the author’s direction. The author reviewed the work and takes responsibility for its content.

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